

02LQD Specification and Simulation of Digital Systems

2023/06/21

1. Combinational circuit design (2 points)

8-4-2-1 is a self-complementary, biased and redundant code. Each decimal digit is expressed on 4 bits. The following table matches BCD and 8-4-2-1:

BCD	8 4 -2 -1
0000	0000
0001	0111
0010	0110
0011	0101
0100	0100
0101	1011
0110	1010
0111	1001
1000	1000
1001	1111

Design in VHDL a BCD to 8-4-2-1 transcoder implementing the above table according to the following description styles:

- dataflow using conditional signal assignments (WHEN ELSE statements)
- dataflow using selected signal assignments (WITH SELECT WHEN statements).

2. FSM design (12 points)

In digital communication, a special synchronization pattern, known as a *preamble*, is used to indicate the beginning of a packet. For example, the Ethernet II preamble includes eight repeating octets of "10101010". Design a synchronous circuit that generates the "10101010" pattern. The circuit has a 1 bit input signal, *start*, and a 1 bit output signal, *data_out*. When *start* is '1', the "10101010" output will be generated in the next eight clock cycles. Reset is synchronous.

Design a **Mealy** FSM with the above specified inputs, outputs and behavior:

- Draw the state diagram (4 points)
- Design the FSM resorting to the VHDL behavioural description style with **1 process** with the same timing behaviour as the 2-process description (4 points).
- Design the FSM resorting in Verilog with **2 always blocks** (4 points).

3. FSM-D design (16 points)

Pell numbers are recursively defined as follows:

$$P_n = 2P_{n-1} + P_{n-2} \quad n > 1$$

$$P_0 = 0$$

$$P_1 = 1$$

The first 8 Pell numbers are thus: 0 1 2 5 12 29 70 169.

Design a FSM-D to **iteratively** compute Pell numbers of order *n*, where *n* is on 3 bits. Computation starts when a *start* signal is asserted for one clock cycle. When computation is over the *ready* signal is asserted for one clock cycle. Reset is asynchronous. The output is *P*. Define the appropriate size for *P*. The output *P* makes sense only when *ready* is 1.

Steps:

3. Define a proper size for the output *P* (2 points)
4. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (6 points)

5. Design a FSM-D:

- create 2 processes for behavioural datapath description from the HLSM state diagram (4 points).
- create 2 processes for behavioural controller description from the HLSM state diagram (4 points).

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

no later than Saturday June 24 11.59 pm

- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.